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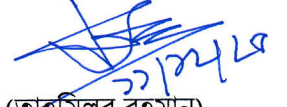
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বিষয়: “নেট মিটারিং গাইডলাইন-২০২৫” এর ইংরেজি অনুবাদ ওয়েবসাইটে প্রকাশ প্রসঙ্গে।

উপর্যুক্ত বিষয়ের প্রেক্ষিতে জানানো যাচ্ছে, “নেট মিটারিং গাইডলাইন-২০২৫” এর ইংরেজি অনুবাদ ০৯/১২/২০২৫ তারিখে যথাযথ কর্তৃপক্ষ কর্তৃক অনুমোদিত হয়েছে।

০২। এমতাবস্থায়, “নেটমিটারিং গাইডলাইন-২০২৫” এর ইংরেজি অনুবাদ বিদ্যুৎ বিভাগ এবং এর আওতাধীন সকল দপ্তর/সংস্থা/কোম্পানির ওয়েবসাইটে প্রকাশের জন্য নির্দেশক্রমে অনুরোধ করা হলো।

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অনুলিপি: সদয় জ্ঞাতার্থে (জ্যেষ্ঠতার ক্রমানুসারে নয়):

০১। উপদেষ্টা মহোদয়ের একান্ত সচিব, বিদ্যুৎ, জ্বালানি ও খনিজ সম্পদ মন্ত্রণালয় (উপদেষ্টা মহোদয়ের সদয় অবগতির জন্য);

০২। সচিব মহোদয়ের একান্ত সচিব, সচিবের দপ্তর, বিদ্যুৎ বিভাগ;

০৩। অতিরিক্ত সচিব (নবায়নযোগ্য জ্বালানি) মহোদয়ের ব্যক্তিগত কর্মকর্তা, বিদ্যুৎ বিভাগ;

(তাহমিলুর রহমান)
সিনিয়র সহকারী সচিব

NET METERING GUIDELINE – 2025



Sustainable and Renewable Energy Development Authority (SREDA)
Power Division
Ministry of Power, Energy and Mineral Resources
Government of the People's Republic of Bangladesh

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1 Background

Electricity is essential for the socio-economic progress and for improving the living standard of the people. Bangladesh has already achieved 100% electrification. To ensure universal electricity access and energy security, the Power Division has adopted energy diversification as one of its key strategic approaches. Under this strategy, initiatives have been taken to generate environment-friendly electricity from renewable resources along with conventional fossil fuels. A significant increase in the use of renewable energy is also one of the core objectives of the UN Sustainable Development Goal 7 (SDG 7) . Besides, 20% of total electricity demand from renewable sources by 2030 and 30% by 2040 is set as target as per the direction of the Renewable Energy Policy, 2025 .

The main sources of RE are – solar energy, wind energy, hydro, biogas, biomass, geothermal, wave and tidal energy. Due to its geographical location, solar is the most prospective renewable energy source in Bangladesh. A total of 1,350 MW of electricity generation capacity has been established so far by utilizing solar energy, majority of which come from solar parks, solar home systems, rooftop solar, and solar irrigation programs. But since more than 2 acres of land is required to generate 1 MW of electricity from solar energy, it has been difficult to arrange vast stretches of land to build large scale solar power plants. Therefore, the prospect of generating electricity by installing solar systems on unused rooftops of various grid-connected facilities—such as residential buildings and industrial establishments—is being given serious consideration. Installing rooftop solar systems will significantly increase the contribution of renewable energy to overall power generation. Moreover, introducing incentive schemes for on-grid electricity consumers to install such systems will further encourage solar-based renewable energy production.

To encourage renewable energy-based distributed generation, initiatives have been taken to introduce the net metering scheme. Under the net metering system, consumers use the electricity generated from renewable energy systems installed on their own premises and export the surplus electricity to the distribution grid. The exported electricity is credited against the consumer's monthly electricity bill, and any remaining credit is adjusted in the following month. This process reduces electricity costs for consumers. Currently, over 50 countries worldwide, including several SAARC member states, have adopted the net metering scheme. So far, in accordance with this guideline, more than 3,763 net-metered rooftop solar systems have been connected to the six electricity distribution utilities' network across the country, with a total combined capacity of 185 megawatts. As it reduces electricity expenses, consumers are likely to be encouraged to adopt the system. Under these circumstances, the Power Division has formulated and updated the “Net Metering Guidelines – 2025” to facilitate on-grid electricity consumers in availing net metering benefits through the installation of renewable energy systems. Based on implementation experience, the guidelines will be revised and refined as necessary in the future.

1.1 Objective of the Guideline

This guideline has been formulated for the expansion of renewable energy and to utilize the electricity generated at consumer's premises under the net metering system.

2 Definitions and Operating Procedure

2.1 Definitions

Unless there is anything repugnant in the subject or context, the terms used in this guideline shall express the meanings specified below:

Term	Definition
AC	Alternating Current
BEPZA	Bangladesh Economic Zones Authority
BEZA	Bangladesh Export Processing Zones Authority
Billing cycle or billing period	The time period for which a consumer's electricity bill is prepared by the distribution utility.
BNBC	Bangladesh National Building Code
BPDB	Bangladesh Power Development Board
Bulk Tariff	The rate determined by the Commission at which the electricity distribution utility generally purchases electricity from the bulk supplier (BPDB).
Capacity of the renewable energy-based distributed generation system	The rated output AC capacity (VA) of the renewable energy-based distributed system. The aggregate AC output (VA) of the inverters, in case of a solar PV system
CAPEX Model	A model under which the consumer installs a renewable energy system through self-investment and self-management and enjoys the benefits of net metering.
Commercial consumer	A consumer categorized and connected as a commercial consumer by the distribution utility
Commercial Operation Commencing Date (COD)	The date of signing the agreement under clause 3.7(l) following the verification of electricity generation from the renewable energy system installed in accordance with the guideline under clause 3.7(m).
Commission/BIRC	Commission established under Bangladesh Energy Regulatory Commission Act, 2003 (Act No. 13) or any subsequent amendment order thereof.
Connection Point	The point where the renewable energy system is connected to the consumer's internal network.

Consumer	Any individual or organization who is connected as electricity customer by the utility.
Contract/Final Agreement	The bilateral contract signed between the consumer and the respective distribution utility under the CAPEX model, and the tripartite agreement signed among the consumer, the respective distribution utility, and the OPEX investor (third party) under the OPEX model, in accordance with Annex-4 of this guideline, for the implementation of net metering operation following the installation of solar or other renewable energy systems.
Credit Unit	For each billing period, the net exported electricity, calculated using the imported units, exported units, and the previously accumulated (credit) units in accordance with the guideline, at the end of the prior billing period — which shall be zero or a positive value.
DC:AC Ratio	The ratio of the total capacity of solar modules (in kilowatt-peak or megawatt-peak under Standard Test Conditions, representing DC capacity) to the system's AC capacity. This ratio generally depends on system design and the solar irradiance at the specific location.
Distribution Area	The geographical area specified under the license of a distribution utility for electricity distribution.
Distribution network	A distribution system comprising of electrical lines, cables, switchgears, and associated equipment and capable of supplying electricity at 33 kV or lower voltage levels
Distribution System Maintenance Charge	The charge payable for the services provided by the distribution utility in facilitating the use of solar or renewable energy, which is determined as a fixed percentage of the net exported electricity during the consumer's billing period.
Distribution utility	An authority licensed by the Commission to distribute electricity within its designated service area.
Electricity Bill	The monthly or supplementary bill issued by the distribution utility in favor of a consumer.
Electricity Import / Consumption	The electricity drawn by a consumer from the distribution utility's network.
Electricity Supply/Export	The electricity generated from renewable energy sources and supplied by the consumer to the distribution utility's grid.
Eligible consumer	An electricity consumer who abides by all the rules and regulations of the utility and is in compliance with the terms and conditions of Section 3.3 of this guideline.
Extra High Voltage	132,000 volt or 230,000 volt
Grid Code	"Bangladesh Energy Regulatory Commission (Electricity Grid Code) Regulations, 2023" or its latest version

Grid System Maintenance Charge	The service charge payable to the national grid operator, as specified in the guideline, for facilitating the use of electricity generated from solar or other renewable energy sources by the consumer — determined as a fixed percentage of the net exported electricity from the distribution utility to the national grid.
Guideline	Refers to the most recent updated version of the “Net Metering Guideline – 2025”
High Voltage/pressure	33,000 volt
Industrial consumer	A consumer categorized and connected as an industrial consumer by the distribution utility.
Interconnection point	The point of connection between the utility and the consumer
kV	Kilovolt or 1000 volt
kW	Kilowatt
kWh	Kilowatt hour
kWp	It refers to the peak power — that is, the maximum direct current (DC) power that can be produced under Standard Test Conditions (STC).
LCOE	An index that represents the average cost of electricity generation per unit (kilowatt-hour) over the lifetime of a solar or renewable energy system. It is calculated by dividing the total lifetime cost of installation, maintenance, financing, and operation by the total amount of electricity generated during that period.
Licensee	An individual/entity that has obtained a license under the Bangladesh Energy Regulatory Commission Act, 2003.
Low voltage	230/400 Volt
Maximum demand	The maximum level of the simultaneous power demand of all the electrical equipment and system of a consumer’s installation expressed in kW or kVA units.
Medium Voltage/Pressure	11,000 volt
MW	Megawatt or 1000 kilowatt
MWp	1 MWp = 1,000 kilowatt-peak
Net Energy Metering	A mechanism whereby an eligible consumer installs a renewable energy system primarily for his own use and the excess energy may be exported to the grid, for which credit is to be received that may be used to offset part of the electricity bill issued by the Utility for the applicable billing period.

Net Exporter	A situation in which, during any billing period, the amount of electricity exported by the consumer is more than the amount of electricity imported.
Net Importer	A situation in which, during any billing period, the amount of electricity exported by the consumer is less than the amount of electricity imported.
Net meter	An energy meter capable of recording both import and export of electricity.
Net Zero	A situation in which, during any billing period, the amount of electricity imported and exported by the consumer is equal.
OPEX Contract	A bilateral agreement between the OPEX investor and the electricity consumer prior to applying to the distribution utility for installation of a solar or renewable energy system under the OPEX model, in accordance with Annex-6 of this guideline.
OPEX Investor	An organization or company that carries out an OPEX agreement for the installation of a solar system under the OPEX model of this guideline.
OPEX model	A model under which a renewable energy system is installed at a consumer's premises by an entity other than the consumer, under a tripartite agreement among the utility, the consumer, and the OPEX investor. The system's generated electricity is then supplied to the consumer.
PCC	The point or busbar within the consumer's premises where electricity generated from renewable energy sources — after passing through the RE Accounting Meter — is connected with the utility's supply (refer to Sample Diagram 5).
pf	Power Factor
PFI	Power Factor Improvement Unit
Prosumer	The electricity consumer who produces and consumes electric energy.
RE	Renewable Energy
RE Accounting Meter	The energy meter, as described in clause 3.6(h) of this guideline, used to measure the total electricity generated from a renewable energy system installed through an agreement.
Renewable energy generator	Machine or equipment that converts renewable energy into electrical energy
Residential consumer	A consumer categorized and connected as residential consumer by distribution utility

Rooftop	Any empty space on the roof, façade, shed of any building, parking lot, garage, boundary wall of any factory/industrial building, the premises or empty space within the boundary of the building/industry, which is under the control of the consumer, or any suitable land where Utility meters exist or any other suitable area accepted by the utility.
Rooftop solar PV system	The solar PV system installed on the consumer's rooftop that can convert sunlight directly into electrical energy through photovoltaic technology.
Sanctioned load/ demand	The amount of maximum power in kW/kVA/MW/MVA or horsepower approved by the distribution utility for supply to the consumer.
Self-Consumption Meter	An optional energy meter used to measure the total amount of electricity consumed by the consumer.
Settlement period	The three-month period at the end of which a consumer receives payment for their accumulated credit in kilowatt-hours.
Settlement Unit	The accumulated credit units of the consumer at the end of the settlement period.
SID	A unique identification number assigned by the distribution utility after the execution of the final agreement, upon entry of system data into the national renewable energy database. This unique number shall be used for all references related to the system after installation.
Solar Accounting Meter	The energy meter, as described in clause 3.6(h) of this guideline, used to measure the total electricity generated from a solar system.
SREDA	Sustainable and Renewable Energy Development Authority
STC	Standard Test Conditions [Irradiance- 1000W/m ² , Cell Temperature- 25°C, Air Mass - 1.5]
Tariff Structure order	An order issued by the Commission to the licensee authorized for electricity distribution.

2.2 What is Net Metering and How It Works?

Net Energy Metering (NEM) is a system in which a prosumer or OPEX investor connects his/her solar or renewable energy system to the distribution grid. As a result, the consumer supplies excess electricity generated from his/her system to the national grid and subsequently draw electricity from the grid at the prescribed tariff rate. At the end of each billing cycle, the consumer is required to pay for the net imported electricity only. If the amount of exported electricity exceeds the imported amount, the surplus is carried forward as a credit. At the end of the settlement period, the consumer receives payment from the distribution utility for the credited electricity units.

Under the Net Energy Metering (NEM) process, a bi-directional meter records electricity flow in both directions separately. This enables the measurement and adjustment between the electricity imported from the grid by the consumer and the electricity exported to the grid from the solar PV system or any other renewable energy source, in compliance with the provisions of this guideline. The import and export data are stored in the meter and transmitted to the central billing server of the distribution utility. For each billing period, the electricity bill is prepared based on the net electricity consumption, calculated by subtracting the exported electricity and the previously credited units from the total electricity imported from the grid, as described in the guideline. If the electricity supplied from the grid exceeds the total of the electricity exported from the renewable energy system and the previous credit, the consumer is required to pay only for the net imported electricity. Conversely, if the electricity exported from the renewable energy system and the previous credit exceed the electricity imported from the grid, the surplus is credited to the consumer's account in kilowatt-hours and carried forward to the next billing period. In such a case, the consumer does not have to pay any energy charges for that month.

At the end of the settlement period, the prosumer receives payment from the distribution utility for all the credits accumulated in kilowatt-hours, calculated at the rate specified in this guideline, after which the credit balance is reset to zero and begins a new cycle. The operation of an ideal Net Metering System is illustrated below through a PV system-based net metering setup:

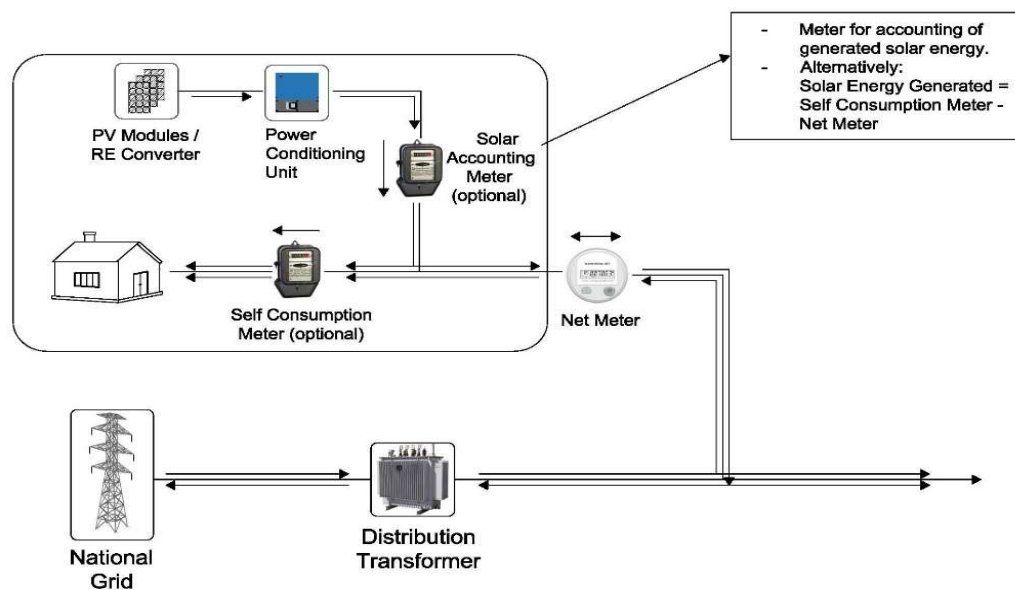


Figure 1: An Ideal Net Metering System

In this context, the Levelized Cost of Electricity (LCOE) for electricity consumed by the customer depends on several factors, including the consumer's tariff category, the type of renewable energy technology used, the installed capacity, and the limitations in power transmission. During the establishment of such an electrical interconnection, the prosumer must comply with all technical specifications and safety requirements prescribed by the competent authority to ensure safe and standardized interconnection.

In accordance with this guideline, renewable energy systems under the Net Metering mechanism can be installed under two models — the CAPEX model and the OPEX model. Under the CAPEX model, the consumer installs a renewable energy system at their own investment and under their own management, thereby availing the full benefits of Net Metering. Conversely, the OPEX model refers to a system whereby a renewable energy facility is installed by a legally authorized third party under a tripartite agreement executed between the utility, the consumer, and the OPEX investor. The electricity generated from the system is supplied to the consumer as per the terms of that agreement. If the electricity consumer is unwilling or unable to install a solar system under the CAPEX model, they may, following due process, engage an OPEX developer company by determining the tariff and necessary conditions, and execute a bilateral agreement allowing the installation of the solar system under the OPEX model. Such bilateral agreement shall be referred to as an OPEX Agreement, which must be concluded prior to submission of the application to the distribution utility. Under this model, the electricity consumer bears no financial or operational risk; all installation, operation, and maintenance activities related to the solar system are undertaken by the OPEX investor company. The consumer purchases electricity generated from the solar system at the rate and under the conditions specified in the contract. The OPEX tariff is generally lower than the retail tariff of the distribution utility.

The success of the Net Metering system fundamentally depends on the integrity of the interconnection process, through which electrical energy is supplied to the distribution grid in compliance with the statutory safety provisions under applicable laws. Although interconnection standards are generally presented separately from the measurement parameters of the Net Metering Guidelines, their significance within the Net Energy Metering framework is of utmost importance.

3 Net Metering Guidelines

3.1 Eligibility of Applicants

An electricity consumer shall be considered eligible for Net Metering subject to fulfillment of the following conditions:

- A. He shall have to be a legitimate consumer of the distribution utility responsible for supply of electricity in the area concerned
- B. While submitting the application for a new connection, an applicant may also submit application for net metering in pursuance of this guideline;
- C. The applicant shall not have any outstanding arrears prior to submitting the application for net metering;
- D. Electricity to be produced ONLY from renewable energy sources would be brought under the scope of net metering;
- E. The applicant must either be the owner or have legal permission from the owner(s) or their legal representative(s) for installing the proposed renewable energy system in the premises of the owner.
- F. Roof of any building, parking lot, garage, commercial or industrial establishment or shed owned by the consumer or any empty space or boundary wall outside of the building but within the premises owned by the consumer or any other suitable area accepted by the Utility and where Utility meter exists is fit for installation of renewable energy system for net metering.
- G. The consumer shall consume the electricity generated by the renewable energy system and after consumption supply the remaining amount to the grid;
- H. Under the net metering scheme, in order to supply the electricity generated from renewables to the grid, connection shall be established at the existing electrical line in compliance with the interconnection rules and standards set by the Utility or any other existing authority;
- I. Instructions given in section 3.3 of this guideline shall have to be followed while determining capacity of the renewable energy system.
- J. Consumers who installed rooftop solar systems prior to the issuance of this Net Metering Guideline may apply to convert their existing systems into Net Metering systems. For such legacy systems (subject to proof of installation date through official utility or government documentation), compliance requirements for solar module standards may be relaxed;

- K. For industries and other establishments located within Economic Zones (EZs) or Export Processing Zones (EPZs) that are not direct electricity consumers of the distribution utility where the relevant authority acts as the utility's customer and may have an associated third-party captive generator. this guideline shall apply to the distribution utility, its direct customer, and the OPEX investor (if applicable). In such cases, the distribution utility shall maintain a Net Metering Agreement with the BEZA/BEPZA authority in accordance with this guideline. BEZA/BEPZA may have additional agreements with other stakeholders (e.g., industries, conventional-energy captive generators, etc.), and such inter-party transaction processes and conditions shall be standardized—either in unified or zone-specific form—and published on the official BEZA/BEPZA website.

Note: A prosumer may supply electricity generated from their own system to any other user through a self-constructed distribution line, provided that the supply network of the electricity distribution utility is not used for this purpose. All related expenses must be borne by the prosumer. However, if the recipient of such electricity is a registered customer of any electricity distribution company, prior authorization from that company shall be required.

3.2 Classification of Consumers

Under the purview of this guideline, the following classes of consumers from the BERC published list of retail electricity tariff may apply to the utility concerned for net metering benefits subject to fulfillment of conditions mentioned in section 3.1 of the guideline:

Low tension (LT): 230/400V	Medium tension (MT): 11kV	High tension (HT): 33kV	Very High Tension (EHT): 132/230
(a) LT-A: Residential (b) LT-B: Pumps used for irrigation/agriculture (c) LT-C1: Small industries (d) LT-D1: Educational, religious and charitable organizations, and hospitals (e) LT-D2: Street lights, water pump, battery charging station. (f) LT-D3: battery charging station (g) LT-E: Commercial and office	(a) MT-1: Residential (b) MT-2: Commercial and offices (c) MT-3: Industry (d) MT-4: General (e) MT-7: battery charging station (f) MT-8: Pumps used in irrigation/agriculture	(a) HT-1: General (b) HT-2: Commercial and offices (c) HT-3: Industry	All classes of consumers

In future, if BERC revises the consumer categories in the retail electricity tariff schedule, the corresponding revised consumer categories shall likewise be applicable.

3.3 Capacity of Renewable Energy Systems and Limits on Export of Electrical Energy

The maximum admissible capacity of the renewable energy system and the limit on export of electrical energy shall depend on the nature of the consumer and the pattern of use. Owing to technical constraints, the following initial conditions have been applied for determining system capacity and export limits:

- A. Any single-phase or three-phase consumer shall be deemed eligible for Net Metering subject to compliance with Clauses 3.1 and 3.2;
- B. The AC output of the renewable energy converter must not exceed the consumer's sanctioned load;
- C. For medium to extra-high voltage consumers, the capacity of the renewable electricity generation system installed under Net Metering must not exceed 80% of the rated capacity of the transformer or the aggregate capacity of the parallel transformers;
- D. If a consumer intends to install a renewable energy system with capacity exceeding the contracted load limit, a separate application shall be submitted to the concerned distribution entity. If deemed appropriate, the distribution entity shall seek the opinion of the Grid Stability Committee (as referenced in Annex-9). Based on the Committee's recommendation, the Power Division shall issue a decision.

Based on experience gained from future deployment of Net Energy Metering systems, the Power Division, Ministry of Power, Energy and Mineral Resources may redetermine permissible capacities.

3.4 Energy Accounting and Settlement

The following procedure shall be followed in case of energy accounting and its settlement:

- A. For each billing period, the concerned distribution utility shall prepare and send the electricity bill to the consumer in the format of Annex-7 (excluding pre-payment/smart meters). In case of pre-payment/smart meters, Annex-8 shall be followed. After adjustment under Clause 3.5, any credited units (kWh) remaining shall be carried forward to the next billing period; any net import shall be billed to the consumer. At the end of the settlement period (every three months), if any kWh credit remains, the utility shall settle the amount with the consumer in accordance with this guideline;
- B. Each bill prepared by the utility for a billing period shall separately state:
 - i. Quantity of electricity received (imported) from the utility's grid;
 - ii. Quantity of electricity sent (exported) to the distribution grid from the installed renewable energy system;

- iii. Previous month's credit units, the billing units for the current month as calculated under the guideline, the credit units calculated for the current month, and, in the final month of a settlement period, the settlement units and their computed value;
- C. A prosumer shall be considered to be a net importer if the net amount of energy received (imported) is more than the net amount of energy sent (exported) by him during a billing period. In that case, the distribution utility concerned shall bill that prosumer for his/her energy use after making adjustments with credits of the previous month of the same settlement period.
- D. A prosumer shall be considered to be a net exporter if the net amount of energy received (imported) is less than the net amount of energy sent (exported) by him during a billing period. In that case, the additional amounts of kilowatt- hour energy shall as credits add to the accumulated credits (if any) from the previous billing period of the same settlement period and be carried forward to the next billing period. In this case, the prosumer shall have to pay the bill for electricity demand charge, meter charge and all other fixed charges.
- E. Energy accounting shall be according to the tariff specification procedure described in section 3.5 of this guideline.
- F. Unadjusted credits may be carried forward up to the settlement period. The settlement period shall end quarterly (i.e., after March, June, September, and December). If, at settlement, no credit units remain, the utility shall have no payment obligation to the consumer, and the consumer shall pay the utility's bill as usual;
- G. Where credit units remain at the end of the settlement period, the utility shall pay the prosumer at the tariff determined under Clause 3.5. Payment shall be made to the bank account specified in the application/contract (BEFTN/NPSB) or to a mobile financial services account (e.g., Nagad/bKash/Rocket), or the prosumer may collect a cheque from the utility, as selected at the time of application;
- H. Under the OPEX model, the electricity consumer shall, pursuant to the mutual contract, pay the OPEX investor for electricity generated by the installed system based on the "RE Accounting Meter." A Self-Consumption Meter may be used to verify meter readings and resolve discrepancies. The consumer's accounting with the distribution utility shall nonetheless follow the processes set out in sub-clauses (a)–(g);
- I. For OPEX systems, if the consumer is a net exporter at the end of the settlement period, the utility shall pay the consumer for credit units; the consumer-OPEX investor reconciliation shall be undertaken in accordance with the OPEX agreement;
- J. If the electricity consumer fails to pay the OPEX installer for two consecutive months, the installer may lodge a complaint with the distribution utility. Upon verification, if found valid, the utility may take action consistent with line disconnection provisions under the Electricity Act, 2018;
- K. If an industrial/commercial consumer's establishment is permanently closed, then, initially, electricity from the OPEX Net Metering system may—with prior approval of the utility—be supplied to another consumer on the same feeder using the utility's distribution lines, accounting for system loss from delivery to draw-off points, or may be supplied directly to the distribution utility at the tariff specified in Clause 3.5(c). For

supply to another consumer, the line loss (on 0.4 kV: per km; on 11 kV: 1% per 5 km; on 33 kV: 1% per 10 km) and transformer loss (2.5%, if applicable) shall be applicable and be adjusted monthly with the new consumer;

- L. If the monthly maximum AC output (kW or MW) from the consumer's rooftop solar system exceeds the approved AC capacity of the renewable system, the consumer shall first receive a written warning. Upon recurrence, the utility may initiate termination of the Net Metering agreement. Such consumers, if net exporters at settlement, shall not receive any value for settlement units;
- M. As an incentive for distribution utilities, 10% of the net exported electricity during a billing period shall be taken as Distribution System Maintenance Charge. The remaining 90% of the net exported electricity shall be credited to the consumer's account.

3.5 Structure of Net Metering

The tariff structure, according to which the utility shall prepare the bill, settle accounts by making proper adjustments or collecting dues at the end of every billing period and settlement period, are described in this section:

- a) For each billing period, the utility shall undertake accounting and settlement of electrical energy in accordance with the Tariff Order issued by the Bangladesh Energy Regulatory Commission (BERC);
- b) Only consumers who have executed agreements as per Annex-4 shall, at the end of the settlement period, receive payment from the distribution utility at bulk rates for settlement units (accumulated credit). For consumers at 33 kV or below, the 33 kV bulk tariff as determined by BERC for the respective utility shall apply; for consumers above 33 kV, the bulk rate applicable to their voltage level shall apply;
- c) For consumers whose consumption is recorded separately for peak and off-peak hours, adjustment of exported units shall first be made against off-peak units of the billing period, and the remaining balance shall be adjusted against peak-hour units;
- d) If BERC revises the tariff structure during any settlement period, the revised tariff shall be applicable for the remaining billing periods of that settlement period and to the annual reconciliation at the end of the settlement period;
- e) Illustrative examples of accounting for net zero, net exporter, and net importer scenarios during a billing period, as well as end-of-settlement accounting and sample bills, are provided in Annex-5 of this guideline.

3.6 Metering Arrangement

Metering shall be undertaken subject to the following conditions:

- (a) The utility shall install at the interconnection point a suitable single/three-phase bi-directional smart meter capable of separately recording imported and exported energy. If an existing meter can meet Net Metering requirements through re-programming, installation of new meter is not necessary;

- (b) The Net Meter shall comply to Annex-2 specifications or be approved by the competent authority (the concerned distribution utility or SREDA). For pre-payment/smart meters, metering and billing shall follow Annex-8;
- (c) For consumers subject to separate peak/off-peak tariffs, a smart meter capable of separate peak/off-peak recording shall be installed;
- (d) The distribution utility shall be responsible for procurement, testing, installation/replacement, maintenance, and meter reading of Net Meters. A prosumer may procure a Net Meter and have it installed through the utility; in such case, the brand and model must be approved by the utility or competent authority (e.g., SREDA). The consumer shall ensure meter security;
- (e) The consumer shall bear the cost of the Net Meter and related accessories;
- (f) Net Meter readings shall constitute the primary basis for accounting. For OPEX billing, the “RE/Solar Accounting Meter” shall also be treated as a primary basis;
- (g) Implementation of Net Metering on pre-payment/smart meters shall follow the instructions in Annex-8;
- (h) For both CAPEX and OPEX models, installation of an “RE Accounting Meter” or “Solar Accounting Meter” is mandatory, with functionality to record monthly maximum AC output (kW or MW) for at least the current month and preceding two months. Where multiple such meters exist due to spatial separation, the aggregate maximum shall be treated as the monthly maximum. A Self-Consumption Meter shall be optional in both models. Mandatory meters shall be tested and sealed by the utility; tampering shall be punishable under the Electricity Act, 2018;
- (i) The utility shall, ensure the use of Automatic Meter Reading (AMR) technology as early as possible;
- (j) For medium and high-voltage consumers whose metering is on the low-voltage side (but who are billed at MV/HV tariffs), transformer losses for exported energy shall be computed at the same rate as for imports;
- (k) The utility may arrange monthly installment supply of grid-tied inverters and Net Meters for consumers;
- (l) Where a utility feeder hosts Net Metering systems with adequate capacity (potential for export to the grid), the utility shall, as required, convert the relevant grid sub-station distribution meters to bi-directional. Such meters shall reconcile electricity exported to the national grid from the utility’s network against electricity imported by the utility from the single buyer (BPDB) during the billing period. In this reconciliation, 5% of net exported electricity shall be treated as Grid System Maintenance Charge and the remainder shall be adjusted. Based on the contribution of variable renewable energy integration (e.g., BESS), a portion of the above 5% shall be transferred from the bulk supplier (BPDB) to the national grid operator at rates determined by the Commission;
- (m) If electricity from sources other than the renewable energy system and the consumer’s grid connection is synchronized within the consumer’s premises, and, at 15-minute intervals, the energy exported to the utility network by the Net Meter exceeds the net flow toward the PCC recorded by the RE Accounting Meter for the same interval, the excess portion shall not be adjustable by the utility for that billing period;

- (n) In the situation described in sub-clause (m), both the Net Meter and the RE Accounting Meter must retain time-of-use (TOU) data at not less than 15-minute intervals for a minimum of three (3) months. The consumer shall submit TOU data for the completed billing period from these meters to the utility, which the utility shall verify;
- (o) If it is necessary to measure electricity flowing from the PCC toward the solar plant (for self-consumption or other purposes), a bi-directional RE Accounting Meter capable of separate import/export measurement shall be used.

3.7 Application Procedure

The eligible consumer, who intend to install and/or connect their renewable energy generation systems with the grid and benefit from net metering shall follow the procedure mentioned in this section.

- a. The eligible consumer shall submit an application to the head of the relevant office of the concerned distribution utility, quoting the consumer number. For OPEX model installations, prior to applying to the utility, and following due process (tender or negotiation, as applicable), the electricity consumer and the OPEX investor shall execute a bilateral OPEX agreement for purchase of electricity generated from solar/renewable sources, in accordance with Annex-6;
- b. At the time of submission, the consumer shall use the application template in Annex-1 to file a hard-copy application, or submit online in a similar format via <https://nem.powerdivision.gov.bd>. All fields of the template must be completed and all supporting documents shall be attached with the application ;
- c. For hard-copy submissions, upon receipt of the fully completed application and supporting documents (including proof of payment, where applicable), the utility shall issue a dated acknowledgement;
- d. For online submissions, the applicant shall receive SMS and email confirmations. The applicant shall also be able to view the application status and next steps online and shall receive SMS and email notifications upon completion of each stage;
- e. The applicant and the concerned distribution utility shall reach agreement regarding installation (where no prior comparable system exists) of the proposed rooftop renewable generation, interconnection, verification and compliance confirmation, approval, and execution of the Net Energy Metering agreement;
- f. Upon successful completion of all actions under sub-clause (e), the utility shall issue a Net Energy Metering Permission in favor of the applicant; otherwise, the applicant shall be informed in writing of the reasons and required actions. For online processing, the email notification shall be considered as permission letter. If any non-renewable system exists at the premises, it shall be explicitly stated in the permission, and the consumer must demonstrate during verification per sub-clause (k) that such system will not participate in Net Metering;
- g. Within ten (10) working days of the submission of the application mentioned in sub-clause (a), the utility shall complete the above-mentioned process and issue the permission letter under sub-clause (f). Where permission under Clause 3.7(f) is issued for a 33 kV or higher-voltage consumer, the utility shall notify the Grid Operator and the

bulk supplier/single buyer (BPDB) concerned zonal office by providing a copy of the permission;

- h. Within the prescribed timeframe, the applicant shall initiate installation of the renewable energy system (where new installation is required) and arrange the necessary interconnection with the assistance of the utility;
- i. The applicant shall complete the Net Energy Metering Checklist as per Annex-3;
- j. Within eight (8) months of receiving the permission letter, the consumer shall complete installation of the renewable energy system along with the required energy meters, submit the completed Net Metering System Checklist to the utility, and apply for verification of compliance with this guideline and the standards set by the competent authority. For online processing, the signed checklist shall be uploaded and apply for going to the next step by ticking the checkbox;
- k. The utility shall inspect and commission the system to ensure conformity of the components and interconnection parameters with the standards in accordance to this guideline and by the utility. If found compliant, the utility shall schedule a date for agreement signing and notify the consumer by SMS/email (for online cases);
- l. The Net Energy Metering Agreement shall be prepared in accordance with the template in Annex-4. For online processing, a PDF copy of the completed agreement shall be available for download by the applicant or the utility. On the scheduled date, the applicant shall execute the agreement with the utility. For the OPEX model, a tripartite agreement shall be executed including the OPEX investor ;
- m. Within fifteen (15) working days of the application submission referenced in sub-clause (j), the utility shall complete all formalities and execute the agreement referenced in sub-clause (l). The date of execution shall be treated as the Commercial Operation Date (COD). If the system is found non-compliant, the applicant shall be informed within this period with the reasons and required actions;
- n. The utility shall upload a scanned copy of the signed agreement referenced in sub-clause (m) to the online application system;
- o. Any cost of modifications required in the utility's system for installation of the consumer's renewable energy system shall be borne by the applicant;
- p. If services are not provided by the utility within the timelines specified in this guideline, the concerned consumer may lodge a complaint with SREDA (online or hard-copy) with appropriate evidence. Similarly, if the consumer considers the rejection reasons or directions provided by the utility to be inconsistent with this guideline, the consumer may file a review application with SREDA (online or hard-copy) with evidence. A Committee constituted by SREDA shall review the complaint or review application and inform the decision to the utility, which shall be treated as final in this matter.

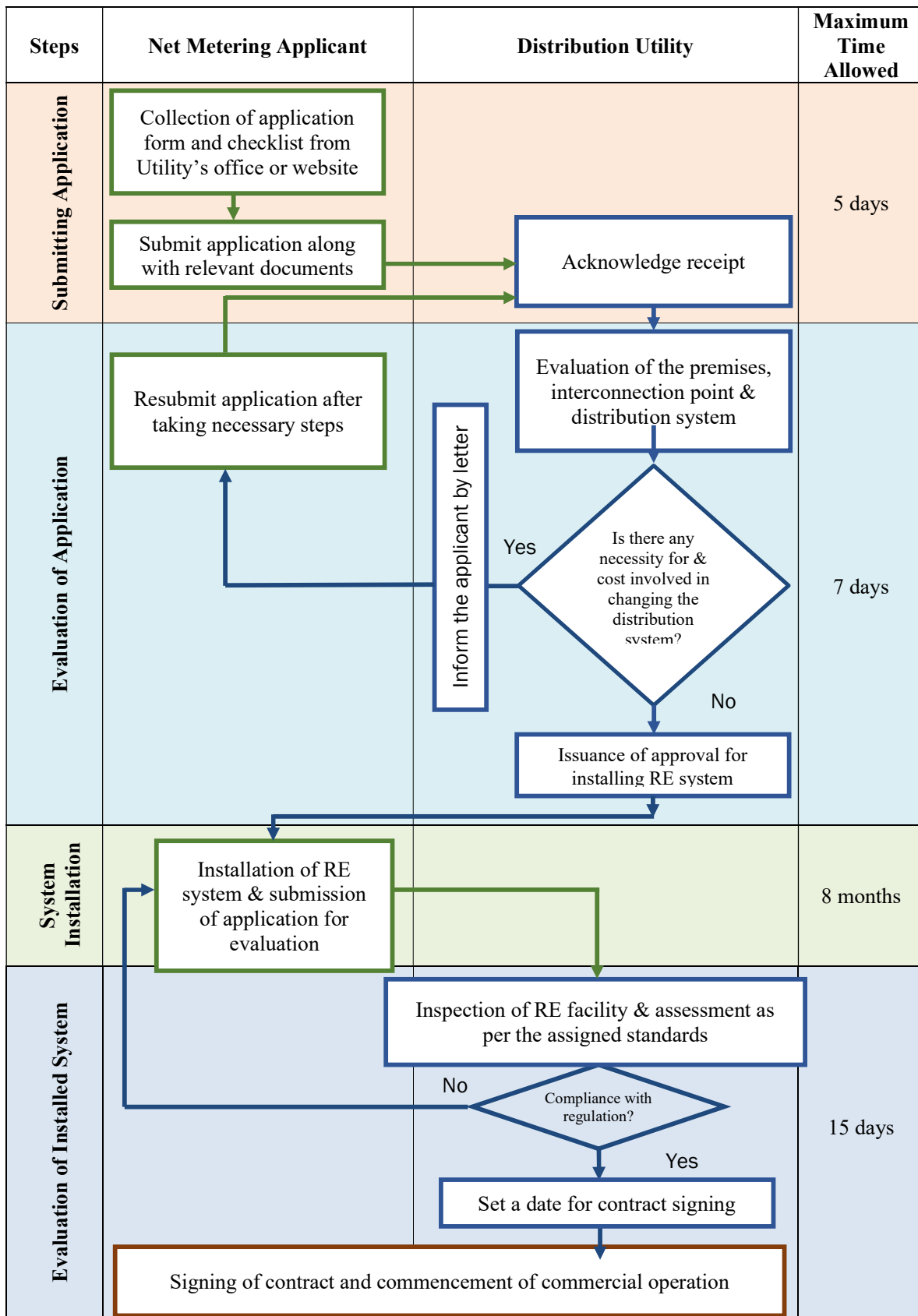


Figure 2: Processing of application for net energy metering

3.8 Approved Equipment

Based on reports issued by testing laboratories enlisted by the Sustainable and Renewable Energy Development Authority (SREDA), clearance for the equipment used in renewable energy systems implemented under Net Metering—namely solar modules, inverters, charge controllers, and Net Meters—shall be provided by SREDA, or a model-wise approved list shall be published on SREDA's website. A separate approval shall be required for each model of such equipment. The following arrangements shall apply:

(a) SREDA shall publish on its website the list of mandatory Bangladesh Standards and the equivalent IEC standards applicable to solar equipment under the Net Metering program, ;

(b) Importers or manufacturers shall import or produce solar modules, inverters, charge controllers, and Net Meters only upon obtaining full compliance (type test) reports from accredited laboratories and certificates from accredited certification bodies against the mandatory Bangladesh Standards or equivalent IEC standards. Accreditation by the Bangladesh Accreditation Board or by Mutual Recognition Arrangement (MRA) signatories to the International Laboratory Accreditation Cooperation (ILAC) shall be acceptable for laboratories. For certification bodies, accreditation by the Bangladesh Accreditation Board or by Multilateral Recognition Arrangement (MLA) signatories to the International Accreditation Forum (IAF) shall be acceptable;

(c) The domestic and international laboratories and certification bodies referenced in sub-clause (b) shall have their accreditation certificates verified and be enlisted with SREDA. Enlisted laboratories and certification bodies shall log in to SREDA's online system to upload their issued test reports and certificates and shall certify full compliance with the mandatory standards;

(d) Until accredited laboratories are established domestically for solar equipment testing, SREDA shall enlist suitable domestic laboratories for sample testing and verification against standards, and shall prescribe sample-testing parameters drawn from the mandatory standards. When accredited domestic laboratories become readily available, sample testing must be conducted at such accredited laboratories;

(e) Upon completion of sample testing as per sub-clause (d) and verification of the full type-test report from an accredited laboratory for the respective model, the testing and verification reports shall be submitted directly to SREDA by logging in to the online system, along with a recommendation for enlistment of the respective model, subject to availability;

(f) SREDA shall enlist the product model on the basis of the type-test (full compliance) certificate and the sample-test report;

(g) Enlisted solar equipment shall have datasheets and descriptions publicly available on the manufacturer's official website.

3.9 Involvement of Distribution Utilities

- a. Following the Renewable Portfolio Standard (RPS), a portion of each distribution utility's total electricity demand shall be generated/imported from renewable energy;
- b. Each distribution utility shall establish a dedicated organizational structure for renewable electricity, with adequate numbers of engineers, officers, and staff appointed/assigned with full responsibility, so that application processing, feasibility studies (where applicable), approvals, and post-installation inspections for rooftop solar Net Metering systems are not delayed;
- c. Pursuant to Clause 3.7 (l) (agreement execution), the distribution utility shall promptly enter the installed system's data into the National Renewable Energy Database (managed by SREDA at <https://ndre.sreda.gov.bd>) and obtain a unique System Identification Number (SID), which shall be used henceforth as the system's identifier in all instances. The distribution utility shall ensure that no duplicate entries are created for the same system;
- d. For each system bearing an SID, the distribution utility shall provide to SREDA, via entries in the National Renewable Energy Database, the monthly/annual electricity generated (in kWh or MWh based on the RE Accounting Meter) and the electricity exported to the distribution network (in the same units based on the Net Meter);
- e. If the annual energy generated by a system installed under this guideline is 35% less than the expected quantity, the distribution utility shall provide the consumer with advisory guidance; if it is below 50%, the distribution utility shall formally request the consumer in writing to undertake remedial measures. A summary of such systems shall be communicated by the distribution utility to SREDA and the Power Division at the end of each settlement period;
- f. Each utility under the Power Division may identify prospective consumers within its jurisdiction (e.g., universities, hospitals, schools/colleges and other educational institutions, shopping malls, prisons, industrial facilities, etc.) and may undertake activities as an investor under the OPEX model;
- g. Distribution utilities shall regularly monitor the power quality of their feeders.

3.10 Capacity enhancement of stakeholders

- a. To promote the expansion of renewable energy through the Net Metering system, the Power Division shall develop skilled human resources by providing necessary training programs.

4 Interconnection Requirements

4.1 Description of the Renewable Electricity Generation System under Net Metering

4.1.1 Feeding method

To reduce the amount of electricity imported from the utility, a consumer may decide to install a renewable electricity generation system under the net metering scheme. A diagram illustrating such an electricity interconnection arrangement is shown below:

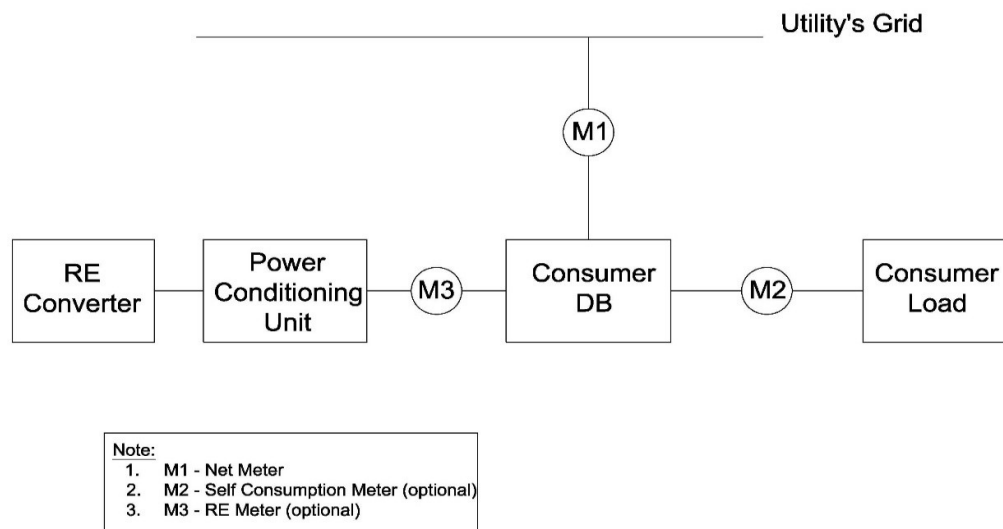


Figure 3: Schematic diagram of indirect connection with the grid

In the above diagram, the amount of electricity consumed from the grid and the amount of electricity exported to the grid shall be measured by M1, the total electricity consumption by the consumer shall be measured by M2 and the electricity generated by the renewable energy generation system shall be measured by M3. In the case of net metering, M1 must be bi-directional and capable of recording Time-of-Use (ToU) readings. If the rooftops within the consumer premises are located at greater distances from one another, considering safety and technical aspects, the AC outputs of the solar systems installed on the individual rooftops may be connected to their respective Consumer Distribution Boards (Consumer DBs). In such cases, there will be multiple M3 meters, and the cumulative total of their readings shall be considered as the overall M3 reading.

4.1.2 Equipment standards

Major components to be installed in Rooftop Solar PV and other Renewable Energy System shall comply with the national standards in terms of design, operation, maintenance and environmental standards. In addition, the grid tied inverters or other renewable energy converters shall comply with the latest edition of IEC 61727 (at present IEC 61727:2004) or

latest edition of IEEE 1547 (at present IEEE 1547:2018) or equivalent SREDA approved standards for Utility interface.

4.1.3 Connection types

Two types of connections may be specified according to the output voltage level of the inverter. The connection types are presented below with corresponding schematics:

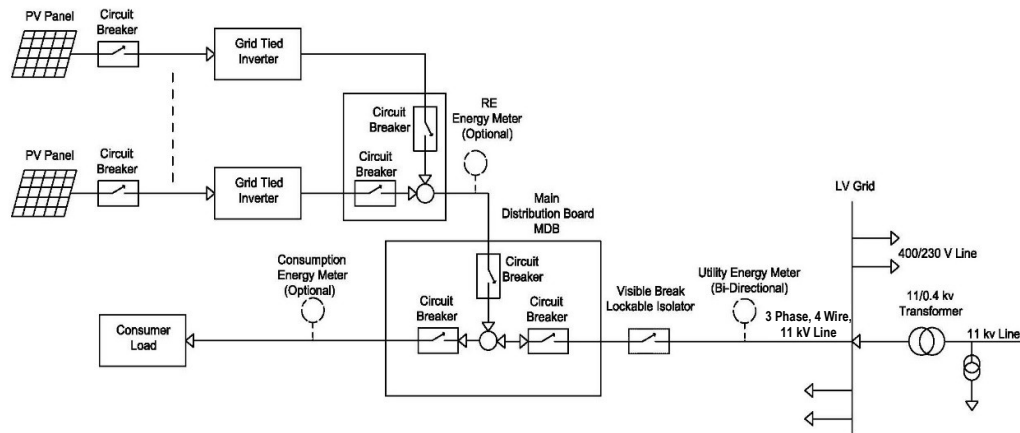


Figure 4.1: Type A electrical connection [Low Voltage Consumer]

Type A is applicable for Utility's consumers with connection to LV network. RE/PV connection point shall be done at the consumer's DB/MSB/SDB. In case of three phase consumers, only three phase inverters can be used in this system.

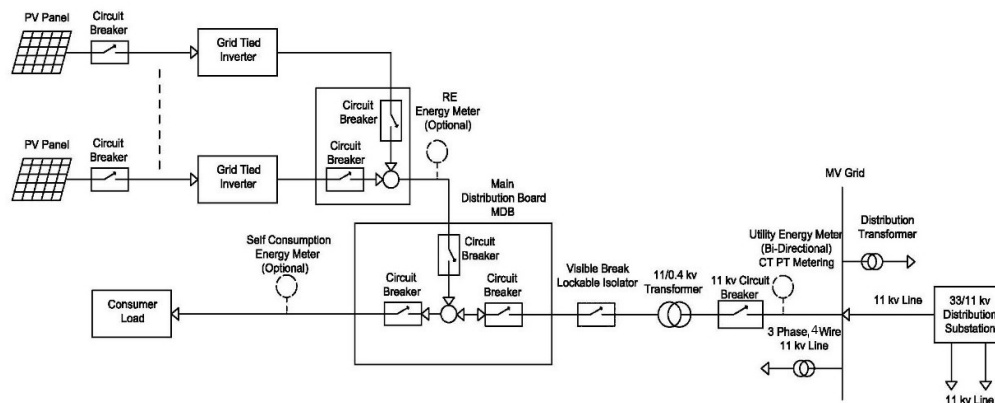


Figure 4.2: Type B connection [Medium and High Voltage Consumer]

Type B connection is applicable for Utility's consumer with connection to MV network. RE/PV connection point shall be done at the consumer MSB/ DB / SDB. In case of three phase consumers, only three phase inverters can be used in this system. This diagram with some modifications shall be applicable for the very high voltage consumers.

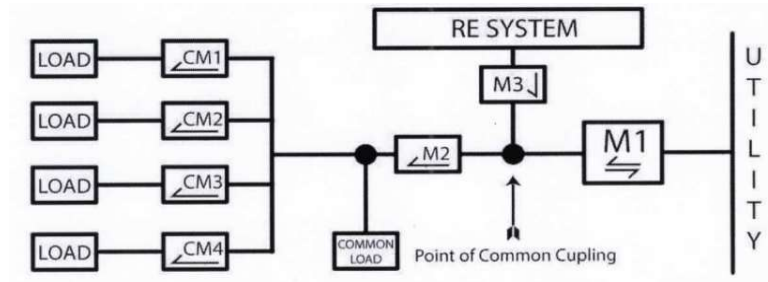


Figure 5: Type C connection [special type of Low Voltage consumers]

For consumers having the distribution Utility's Mother Meter and Sub Meter connected to the LV network shall have Type C connection. Circuit breakers and isolators similar to those shown in Figure 4 shall be applicable. Three-phase inverters shall be used for such connections. Medium and high voltage consumers may also be connected in this way after carrying out all the necessary changes. For determining the system capacity, cumulative supplied load of all the meters shall be taken into consideration.

4.2 General Conditions of Interconnection

If the interconnection requires any enhancement of the existing Utility supply infrastructure such as cable, fuse, switchgear, and transformer and protection scheme, the consumer shall bear the cost of such modification.

The quality of the power at the point of interconnection shall not be lower than the existing quality of supply as mentioned in subsection 4. Quality of supply is measured as per the standards of voltage, flicker, frequency, harmonics and power factor, as specified by relevant and concerned authorities. Deviation from these standards represents out-of-bounds conditions and the net metered system shall be able to sense the deviation and disconnect itself from the distribution network accordingly.

Prosumers connected under this guideline using renewable energy sources must comply with the applicable Distribution Code and Grid Code according to their voltage level.

The net metered consumer shall have to meet the following requirements in order to ensure that the interconnection does not adversely affect the quality of supply.

4.2.1 Normal voltage operating range

Acceptable limit of voltage in case of supplying electricity to the distribution grid from renewable energy source under the net metering scheme is as follows:

- i. Renewable energy systems connected with a low-voltage (LV) interconnection shall operate within the voltage range specified in **Table 1**.

Table 1: Normal operating condition at Low Voltage interconnection

Nominal voltage [V]	Steady state voltage limit
400	-15% to +10%
230	-15% to +10%

- ii. Renewable energy systems connected with a medium and high voltage interconnection shall operate within the voltage range specified in **Table 2**.

Table 2: Normal operating condition at Medium and High Voltage interconnection

Nominal voltage [kV]	Steady state voltage limits
11	-15% to +10%
33	-15% to +10%

- iii. The range of voltage and other issues mentioned in the national grid will be followed in case of very high voltage i.e. at 132 kV or 230 kV voltage level.

4.2.2 Voltage fluctuation

Due to the inherent variability of renewable energy sources, the electricity generation from renewable systems under net metering may fluctuate. As a result, when electricity from renewable sources is supplied to the distribution network, it may cause voltage variations at the interconnection point and within other buses of the grid.

The maximum allowable voltage fluctuation due to such variability shall be 6% for low, medium, and high voltage systems. If the fluctuation exceeds this limit, electrical equipment at the utility or consumer end may overheat. Therefore, an appropriate voltage regulation mechanism must be adopted to minimize such fluctuations when necessary.

4.2.3 Power Factor of Electricity Generated from Renewable Energy

The power factor is defined as the ratio of applied active power to apparent power. The inverter configuration of the renewable energy system shall be set (with the assistance of a Power Factor Improvement (PFI) unit, if required) so that the monthly average power factor recorded in the utility billing meter remains within the limits prescribed by the Commission for that consumer category.

- i. When the inverter operates at a load exceeding 20% of its rated output, the renewable energy system must be capable of operating at any leading or lagging power factor greater than 0.85. If a smart inverter is used, it shall automatically adjust to ensure that voltage rise due to power factor remains within allowable limits.
- ii. The required plant power factor shall be determined during the technical assessment phase.

4.2.4 Reactive power compensation

For renewable electricity generation systems under net metering operating at unity power factor ($\text{pf} = 1$), all reactive power for the consumer load shall be imported from the utility, while real power will be a combination of on-site generation and imported utility power. Since self-generation typically requires a higher proportion of reactive power, the utility tariff meter may show a low power factor reading. To avoid this, consumers must comply with the instructions outlined in Section 4.2.3. In OPEX model solar systems, the required amount and conditions for reactive power to be drawn from the plant shall be explicitly included in the OPEX agreement.

4.2.5 Injection of direct current

The amount of DC current injected by the PV system into the Utility interface shall under any operating conditions never exceed 1% of the rated inverter output current.

4.2.6 Harmonic

The harmonic of a wave is a component frequency of a wave that is an integer multiple of the fundamental frequency. In the presence of non-linear loads (e.g., computer power supplies, electronic devices), alternating current may become distorted due to harmonic frequencies. Harmonics are measured as a percentage of the fundamental frequency or by calculating Total Harmonic Distortion (THD). Excessive harmonics can damage electrical systems and connected loads. To prevent such damage, the following measures must be ensured:

- i. The PV system output shall have low current-distortion levels so that other equipment connected to the Utility system is not adversely affected.
- ii. Total harmonic current distortion shall be less than 5% of the rated inverter output at the cable connected to the interconnection point.

4.2.7 Voltage unbalance

Voltage unbalance is defined as the ratio of negative sequence voltage component and positive sequence voltage component.

- i. Infrequent short duration peaks with a maximum value of 2% over 1-minute duration may be permitted for voltage unbalance.
- ii. When multiple single-phase PV units are installed, the unbalance shall be distributed evenly among the three phases of the power system.
- iii. The unbalanced voltage shall not exceed 1% more than 5 times within any 30-minute period at the terminals of the consumer's installation.

4.2.8 Short circuit level

The distribution utility must ensure that the short-circuit level of the network is appropriate in accordance with the equipment ratings. The maximum sub-transient three-phase symmetrical short-circuit current of the network must remain within 90% of the short-time make-and-break capacity of the equipment. The distribution utility shall be responsible for verifying compliance with this requirement. Table 3 presents the standard equipment rating values commonly used by the utility.

Table 3: Typical equipment rating in distribution network

Nominal voltage [kV]	Rated voltage [kV]	Fault current [kA]
33	36	25
11	12	20
0.4	1.0	31.5

4.3 Protection System

The safety system of a renewable energy generation setup installed under the Net Metering framework must be designed in such a way that it can isolate any faulty section from the properly functioning part of the system when required. The prosumer shall design their protection system to ensure adequate safety based on their operational needs. However, Net Metering customers must comply with the protection requirements imposed by the utility, ensuring that any fault within the renewable energy generation system does not propagate beyond the system boundary.

4.3.1 Smart inverter

The network may become unstable due to voltage fluctuation at the interface of distribution network and RE resulting from increase or decrease of load at the user end. This may happen especially in case of high level of penetration of distributed generation.

Advanced inverters, also known as smart inverters have a number of additional features along with the capability of converting direct current into alternating current. Smart inverters have the capability to assist in retaining grid stability when necessary. These features include:

- Reactive power control
- Active power control
- Grid management

The inverter used in a consumer's system must comply with the **requirements outlined in Sections 4.3.3 to 4.3.9** of this guideline.

4.3.2 Frequency

The distribution utility must maintain system frequency, and the PV system shall operate in synchronization with the utility's frequency. The utility shall operate within a $\pm 1\%$ band of 50 Hz, and the inverter must be capable of generating power within a $\pm 4\%$ frequency range.

4.3.3 Synchronization

Synchronization is the process of being coherent within allowable limits. The energy generation system under the net metering scheme shall have to be equipped with automatic synchronization system. In case of solar PV, synchronization has to be done at the inverter.

4.3.4 Anti-islanding inverter

- i. Anti-islanding or non-islanding inverters are unable to supply electricity in the absence of grid electricity. To ensure the safety of the solar PV system, it shall not be allowed to remain energized during outage of Utility grid (loss of mains). Within 2 seconds of power outage, the prosumer's electricity generating system from the PV system shall have to be automatically disconnected, which is possible through an anti-islanding inverter. The grid-tied inverter used in the net metering system shall have to be capable of anti-islanding as per IEC 62116 (Utility interconnected photovoltaic inverters – Test procedure of islanding prevention).
- ii. Inverters used by the NEM consumer shall provide the following anti-islanding detection techniques:
 - a. Under voltage
 - b. Over voltage
 - c. Under frequency
 - d. Over frequency

It shall also include at least one of the following active anti islanding techniques:

- a. Negative-sequence current injection,
 - b. Impedance measurement,
 - c. Slip mode frequency shift,
 - d. Frequency bias, etc.
- iii. During the commissioning test, the NEM consumer shall have to prove whether the plant has anti- islanding capacity or not.

4.3.5 Inverter fault current contribution:

The inverter fault current is generally regulated by the inverter control. According to IEC 61727 or IEEE 1547, the typical range of short circuit current is between 100% and 200% of the rated inverter current. The net metering consumer shall have to ensure that the inverter used by him meets the requirements of IEC 61727 or IEEE 1547.

4.3.6 Protection schemes:

The fundamental conditions for developing a protection scheme are as follows:

- i. No harm shall be allowed to the distribution network or any other consumer thereof due to any internal fault existing in the renewable energy system installed under the net metering scheme;
- ii. Adequate safety measures must be in place to prevent damage to the renewable system due to faults in the distribution network;
- iii. In addition to the above, the net metering consumer must maintain all supplementary safety measures necessary for both the distribution network and the renewable generation system.

4.3.7 Failure of system protection or control equipment:

In the event of any system malfunction, the renewable energy system installed under the Net Metering scheme must be disconnected from the distribution network. The situations in which the renewable energy system installed under the net metering scheme must be disconnected from the distribution network are as follows:

- i. Failure of protection equipment
- ii. Failure of control equipment
- iii. Loss of control power

4.3.8 Frequency Withstand Capability

The renewable energy system installed under net metering must meet the Frequency Withstand Capability requirements specified in the Grid Code, for both under-frequency and over-frequency events.

4.3.9 Voltage Disturbance

- i. The inverter of the renewable energy system installed under the net metering scheme shall have the technical ability to sense any abnormal voltage according to standards mentioned in table 4. If there is any fluctuation in transformer, wiring or feeder circuit voltage, the inverter shall take measures according to **Table 4**.

Table 4: Voltage disturbance

Voltage at interconnection	Maximum trip time (s)
$V < 50\%$	0.10
$50\% \leq V < 85\%$	2.00
$85\% \leq V \leq 110\%$	Continuous operation
$110\% < V < 135\%$	2.00
$135\% \leq V$	0.05

- ii. The inverters shall remain functional within the range from -15% to +10% of the nominal voltage of the distribution Utility.
- iii. Voltage may be affected due to voltage disturbance caused by transmission network switching and distribution switching on nearby feeder. Therefore, inverters shall have to be able to ride through voltage disturbance bands ranging from 50% to 85% and from 110% to 135%. This shall help stabilize the Utility's system.
- iv. Loss-of-mains shall be indicated by a voltage drop of less than 50%.
- v. Over voltage and under voltage detection shall be provided for all 3 phases.

4.3.10 Utility interface disconnect switch

To ensure the safety of both the system and personnel while performing work on the distribution utility line, a Utility Interface Disconnect Switch must be installed to allow the renewable energy system set up under Net Metering to be disconnected from the utility supply. A standard switch may be used for this purpose, provided that it is manual and lockable. In addition, the switch must possess the following characteristics:

- The position of the switch must be clearly indicated.
- It must be easily visible and accessible to personnel performing maintenance or operational duties.
- When in the “OFF” position, the contact points must be visibly separated

4.4 Safety Requirements

The installation of renewable energy systems under the grid-connected Net Metering framework must adhere to both national and international safety standards. For the safe operation of a grid-connected system, due consideration must be given to the system topology and the earthing (grounding) arrangements.

4.4.1 Operation

- i. In the interest of safety for operating personnel and the public, both the distribution utility and the Net Metering customer must maintain proper coordination. When any work or testing is carried out at the interface/connection point, necessary arrangements must be made to isolate the system from the distribution grid and to ensure proper earthing.
- ii. When any work or testing is performed at the interface between the distribution network and the renewable energy system—including rooftop solar systems under Net Metering—both the distribution utility and the Net Metering customer must strictly comply with all statutory laws, rules, regulations, license conditions, utility safety rules, and the existing Grid Code and Distribution Code of the country to ensure safety.

4.4.2 Labeling

Labels or markings on equipment and associated cables must be placed in clearly visible locations so that operators can easily identify them and exercise caution during operation or maintenance. This is essential because certain parts of the renewable electricity generation system may remain electrically energized.

5 Amendment of the Guideline

The Government may, as and when necessary, amend and refine this Guideline.

6 Annex 1: Application Template

To

.....

.....

i Applicant Information

Consumer category:

Select the system installation model:

□ CAPEX model

□ OPEX model

In the case of a system installed under the OPEX model, the following information about the investing organization must be provided:

Name of the organization/company:	
Registration No.	
Trade License No.	
Address:	

If the applicant (consumer) is an individual:

Name:																								
Address:																								
Nationality:																								
National ID:																								
Passport no. (If non-Bangladeshi):																								
Telephone:															Mobile:									
Email:																								

Alternative contact person:

Name:	
Address:	
Relationship:	
Nationality:	

National ID:																					
Passport no. (If non-Bangladeshi):																					
Telephone:											Mobile:										
Email:																					

If the applicant (consumer) is an entity:

Name of the Organization/Company:	
Registration no.:	
Trade License no.	
Address:	
Mailing Address (if different):	

Contact Person:

Name:																					
Designation:																					
Address:																					
Nationality:																					
National ID:																					
Passport no. (If non-Bangladeshi):																					
Telephone:											Mobile:										
Email:																					

ii Information on the Renewable Energy / Electricity Generation System

2.1 Installation Site Address

Site Address:			
Post code:		District:	
Ownership of site:	☐ Fully Owned	☐ Mortgaged to a bank	☐ Leased property
GPS Location of the Installation Site:			
Latitude*:		Longitude:	

* Latitude and longitude have to be written with 6 (six) digits after the decimal point (e.g.23.039612, 90.401387) to clearly identify the location as shown in the examples.

2.2 Information about installation

Utility & concerned office:																	
Consumer No. / Billing Account No.:																	
Tariff Category:																	
Sanctioned Load:																	
Current Meter No. (if any):																	
Proposed system capacity:	AC (Kw)		DC (Kw _p)		DC : AC ratio												

2.3 Information about the installation premises:

Voltage Level at the Connection Point:	☐ Low Voltage (Single Phase)	☐ Low Voltage (Three Phase)	☐ Medium Voltage (Three Phase-11 kV)	☐ High Voltage (Three Phase-33 kV)	☐ Very High Voltage (Three Phase-132 kV)	☐ Very High Voltage (Three Phase-230 kV)
Voltage at Point of Common Coupling (RE):						
<i>ONLY for Commercial and Industrial Category:</i>						
Low Voltage	Whole Current $\leq 100A$, Fuse Rating:					
	or, CT Rating:					
Medium Voltage:	CT Rating:		PT Rating:			
Renewable Energy System	☐ New Project			☐ Existing Project		
Types of Building (House/ Shop/ Office/ Others etc.)						
Types of Installation:	☐ Rooftop or facades of Building	☐ Car Park or Garage	☐ Others:			
Use of Battery Storage:	☐ Yes			☐ No		
<i>If yes, please provide detailed design:</i>						
Battery Capacity:			Technology			
Brand and Model:			Country of Origin			

Description of Inverter (If applicable)	
---	--

2.4 Technical Self-Assessment [ONLY for Commercial and Industrial Category]

Daytime Peak Demand (09 am to 05 pm)	kWac
Daytime Lowest Demand (09 am to 05m)	kWac
Weekends / Holidays Peak Demand (09 am to 05 pm)	kWac
Weekends / Holidays Lowest Demand (09 am to 05 pm)	kWac

iii Method for Receiving Settlement Payments

The applicant shall select one of the following methods. The account name must match the electricity consumer's name.

☐ Method 1: Bank Account (BEFTN/NPSB)	☐ Method 2: Mobile Banking Account (Nagad / bKash / Rocket)	☐ Method 3: Cheque
1. Account name: 2. Account number (13 digits): 3. Bank name: 4. Branch name: 5. Routing number:	1. ☐ Nagad / ☐ bKash / ☐ Rocket 2. Account name: 3. Account number:	1. Account name:

iv Applicant's Declaration

**To be completed by an (individual) applicant*

I,,
National ID No./ Passport No.:
..... and address:
..... do hereby declare that:

- i. I hereby authorize..... (name of the authorized person) with National ID No./Passport No.:

..... to process all matters related to net metering on my behalf;

OR

I hereby authorize..... (name of the authorized organization) with registered address at to process all matters related to net metering on my behalf; its Registration No.: and name & address of the registering authority;

OR

I, National ID/Passport No. declare myself as the competent person to process all matters related to net metering.;

- ii. I hereby certify that the authorized person or organization (if any) is competent to comply with this Guideline and all instructions issued by the competent authority in this regard;
- iii. I hereby certify that only approved equipment under this Guideline will be installed in the proposed renewable energy/electricity system, and compliance with the stated specifications and national standards will be verified.
- iv. I hereby certify that electricity generated from any source other than renewable energy shall not be net-metered, and compliance shall be ensured as per the Guideline.
- v. I hereby warrant that I have not committed any offense punishable under the Electricity Act, 2018 or any other law or regulations relating to electricity supply and licensing.
- vi. I hereby certify that all information provided herein is true and correct to the best of my knowledge and belief.
- vii. I acknowledge that if any information provided herein is found to be false, the distribution utility reserves the right to forfeit the application fee and to take any action against me as deemed appropriate.
- viii. I hereby undertake to comply with all applicable laws, rules, regulations, and Government instructions relevant to this application;

.....
Signature

7 Annex 2: Detailed Description of the Net Meter

1. If existing unidirectional electric meters cannot be reprogrammed to meet the requirements of this guideline for bidirectional operation, they must be replaced with new bidirectional meters.
2. The existing meter board and related meter wiring (if necessary) must be replaced either with the assistance of the distribution utility or by a technician authorized by the distribution utility. The location where the meter is installed must be accessible to the utility's personnel.
3. The consumer shall bear all costs associated with the interconnection including the cost of meter replacement, supply upgrading, and system connection/ modification.
4. The installed net meter system must have a remote online monitoring facility. Both the net metering consumer and the OPEX investor shall grant online monitoring access to the utility.

(a) Minimum Technical Specifications for Net Metering Measurement:

No.	Technical Parameters	Low Voltage Level Connectivity	
		Whole current meters	CT operated
1	Measured quantities	kWh, kVAh, kVA, Power Factor	kWh, kVAh, kVA, Power Factor
2	Accuracy class	0.5	0.2
3	Import and export capability	Import & Export	Import & Export
4	Communication port/protocol	Optical / RS-232 / DLMS / MODBUS / Wireless	Optical / RS-232 / DLMS / MODBUS / Wireless

Net Metering shall be implemented at the applicable voltage level for the allocated load and tariff category in accordance with the Retail Tariff Order issued under the Bangladesh Energy Regulatory Commission Act. In this regard, the Net Metering customer must comply with the standards and requirements set by the distribution utility.

(b) Prepayment/Smart Meter:

Pre-payment/Smart meter customers must have a metering system capable of recording and storing the amount of electricity imported from and exported to the distribution grid. The meter must also have the capability to perform billing calculations in accordance with this guideline. In this regard, Annex-8 of the guideline must be followed.

8 Annex 3: Net Metering System Checklist

1. Applicant information

Name (<i>Person/ Organization</i>):	
Address:	
Utility account no.	
Meter no.	
Sanctioned load	

a. Contact person:

Name:	
Designation:	
Address:	
Nationality:	
National ID:	
Passport no. (If non-Bangladeshi):	
Telephone:	Mobile:
Email:	

2. Components used

a. Solar PV panels

Manufacturer (<i>Company, origin</i>)	
Brand	
Model No.	
Rated capacity (w_p)	
Standard Compliance: BDS IEC 61215-1:2022, BDS IEC 61215-2:2022, BDS IEC 61730-1:2019, BDS IEC 61730-2:2019 or latest versions	
Description of compliance certificate (mandatory) in support of national standards developed by BSTI (BDS	

IEC 61215 and others)	
Other applicable compliance standards	
Approval No. & reference for use in Net Metering Program	
Number of Panels	
Total installed capacity	

b. Inverters

Manufacturer (<i>Company name, origin</i>)	
Brand	
Model no.	
Rated capacity (wp)	
Standard Compliance: BDS IEC 61727:2020, BDS IEC 62116:2020, BDS IEC 62109-1:2016, BDS IEC 62109-2:2016 or latest versions	
Other applicable compliance standards	
Available and selected operating options for power factor control	
Approval No. & reference for use in Net Metering Program	
Number of inverters	
Installed capacity (<i>Total</i>)	

Note: The list of mandatory standards for solar modules and inverters may be updated periodically. Please refer to the latest version available on the SREDA website (www.sreda.gov.bd).

c. Mounting system

General Information	
Manufacturer (company, origin)	
Type & description (where applicable)	
Type of rooftop and mounting system	
Mounting location	☐ RCC roof ☐ Tin shed ☐ Open space within premises ☐ Others
Rooftop angle(s)	
Design type	☐ Seasonal tracking ☐ Daily tracking ☐ Fixed ☐ Others
Type, material, and load per square meter of mounting structure	
☐ I/We hereby certify that all BNBC and building ownership regulations, as well as technical specifications related to building structures, have been complied with	

d. Cables/ power lines

General description		
PV string cable / power line		
Manufacturer		
Type		
Cross-section		
PV Array Cables / Power Lines (DC)		
Manufacturer		
Type		
Cross-section		
Inverter supply cable / power line (AC)		
Manufacturer		
Type		
Cross-section		

e. Feed-in Management/ communication

General information	
Features	☐ Data collection, storage, circulation ☐ Effective power reduction ☐ Others : _____
Implementation with	☐ Permanent inverter settings ☐ Feed-in management system ☐ Other measures: _____
Manufacturer (company, origin)	
Type & description (where applicable)	
Does the system display effective feed-in levels to network operator? ☐ Yes ☐ No	

3. Information regarding planning and installation

General	
☐ The PV system has been installed in accordance with all existing technical regulations and standards. ☐ The PV system was built with minimal shadowing effect. ☐ A shading analysis was incorporated into the yield forecast if degree of shading is notable.	
Structural information	
☐ For rooftop-mounted systems, load-bearing capacity of the structure has been assessed by a qualified person or firm (with address and phone): ☐ In case of roof-mounted system, the aging condition of the rooftop surface has been assessed by a qualified person or firm (with address and phone): ☐ Anchoring and loading applications have been performed in accordance with manufacturer's instructions or mounting system guidelines. ☐ The fastening of the modules was carried out in accordance with the manufacturer's guidelines.	
☐ If any alternative means has been used for mounting or installation of the system, please specify the details.:	
☐ Roof perforations were carried out in accordance with technical rules and standards.	
Building details:	
Building height	Meter
Wind load zone	
Maximum wind speed for zone	Meter/Second

Cable		Meter
Edge Distance		Meter
Roof Ridge		Meter

Fire Protection Measures

- ☐ Smoke and heat exhaustion systems are fully functional.
- ☐ Firewalls and fire compartments have been considered as per applicable fire protection regulations.
- ☐ Other measures:

Electrical safety, choice and installation of electrical equipment

The equipment and system have been installed in accordance with the general provisions of IEC 60364-1 (DIN VDE 0100), as well as the specific provisions of IEC 60364-7-712:2017 (VDE 0100-712) and VDE-AR-N 4105 (or their equivalent IEC standards). The installation has been inspected in compliance with IEC 62446 (VDE 0126-23). Along with other characteristics, the following features have been found to be present:

- ☐ The cables and power lines of the renewable energy system have been selected and installed in accordance with IEC 60364-5-52 (VDE 0100-520) to ensure that there is no risk of earth faults or short circuits.
- ☐ The cables are securely attached to the frame, not in contact with the roof, and have not been laid over any sharp edges. Appropriate strain relief has been provided at every connection point. The cables located outside the building have been routed through covered cable trays.
- ☐ Adequate circuit breakers have been provided for the DC current circuits in accordance with the manufacturer's instructions, ensuring suitability and sufficient switching capacity.

Lightning and voltage surge protection

Note: It is mandatory for a PV system to have a lightning protection system:

1. Lightning protection system is mandatory for buildings. Is there any lightning protection system in the building?	☐ Yes	☐ No
2. Is there a test report for the lightning protection system?	☐ Yes	☐ No
3. Has the outside lightning protection system been adjusted accordingly?	☐ Yes	☐ No
4. Have distances between lightning protection components been measured?	☐ Yes	☐ No
5. Is the PV system within the protected area of the lightning protection system?	☐ Yes	☐ No

6. Is the PV generator directly connected to the lightning protection system?	☐ Yes	☐ No
7. Has lightning protection equipotential bonding been carried out?	☐ Yes	☐ No
8. Has a Type 1 DC lightning arrester been installed near the PV string cable entry?	☐ Yes	☐ No
9. Has a Type 1 AC lightning arrester been installed at the inverter AC end?	☐ Yes	☐ No
10. Has a Type 1 AC arrester been installed at the building's conduit entry?	☐ Yes	☐ No
11. Has a Type 2 DC surge arrester been installed at the inverter DC side?	☐ Yes	☐ No
12. Has a Type 2 AC surge arrester been installed at the inverter AC side?	☐ Yes	☐ No
13. Has a type 1-2-3 combination arrester been installed at the feed conduit's building entry point?	☐ Yes	☐ No
14. Has equipotential bonding been completed for the mounting structure?	☐ Yes	☐ No
Note: <i>Separate lightning protection = min. 4 mm² (unprotected cable channels). Combined lightning protection = min. 16 mm²</i>		

9 Annex 4: Net Metering Agreement Form

This Agreement is prepared and executed on the ____ day of _____, _____, at _____ (place of execution), by and between **the electricity consumer** named _____, **Consumer** No. _____, under _____ (name of distribution office), **address of the consumer** _____, hereinafter referred to as the **First Party**;

AND

_____ (name of the Utility), _____ (head office address), established under the authority of _____ Act/Ordinance, or registered under the **Finance Company Act, 2023 (Act No. 59)** and operating as a “Distribution Utility” under the **Bangladesh Energy Regulatory Commission Act, 2003**, hereinafter referred to as the **Second Party or Utility**; this term shall include its authorized representatives and successor entities;

AND

Whereas, _____ (name of company), _____ (address), **Company Registration No.** _____, **Registering Authority** _____, has installed a solar system at the consumer’s premises under the **OPEX model**, hereinafter referred to as the **Third Party**;

AND

Whereas, _____ (name of electricity consumer) has completed installation, within his/her/its owned premises located at _____ (location name), of a renewable energy system of _____ kW AC (_____ kWp DC) capacity, connected at _____ **voltage level** to the distribution grid of _____ (name of Utility), in accordance with the Net Metering Guideline;

AND

Whereas, by Order No. _____ dated __, issued by _____ (name of competent authority), approval has been granted for the development and installation of a renewable energy system of ** kW AC (kWp DC)** capacity at the premises owned by _____ (name of electricity consumer), located at _____ address (including

roof and/or principal premises), under _____ (name of policy), to meet own electricity demand;

AND

Whereas, the Distribution Utility has agreed to provide **grid interconnection** for the electricity generated from the _____ kW AC (_____ kWp DC) rooftop solar PV system of the eligible consumer, subject to compliance by the concerned consumer with the terms of this Agreement and with the applicable provisions of the following policy/rule/regulation/code (as amended from time to time):

(List of relevant documents as specified by the Utility)

NOW, THEREFORE, the Parties hereby agree as follows:

1. General Eligibility Requirements

- i. The First Party has fulfilled the conditions set forth in Clause 3.1 of the Net Metering Guideline–2025.
- ii. The First Party and, where applicable, the Third Party shall duly comply with the provisions regarding generation limits and export limits to the grid as governed by Clause 3.3 of the Net Metering Guideline–2025.
- iii. The First Party and, where applicable, the Third Party agree that electricity export to the grid from renewable energy sources shall be in full compliance with Government-regulated rules and any amendments issued from time to time

2. Technical and Interconnection Requirement

- i. The First Party and, where applicable, the Third Party agree that the installed renewable energy system shall conform to the Net Metering Guideline–2025, applicable regulations, the prevailing Grid Code and Distribution Code, and any subsequent amendments and specialized standards and requirements therein.
- ii. The First Party and, the Third Party (where applicable) agree that, prior to connecting the rooftop solar PV system to the Utility's distribution grid, a Utility Interface Disconnect Switch (with both automatic and manual provisions) has been or will be installed; and further agree to provide the Second Party access to and control over the same for repair and maintenance of the distribution system.
- iii. The First Party and, the Third Party (where applicable) agree that, upon loss of grid supply, the installed renewable system shall automatically disconnect, and no power from the said system shall flow into the Utility's distribution system.
- iv. All equipment connected to the distribution system shall be operated in accordance with all relevant provisions specified in this Guideline.

- v. The First Party and, the Third Party (where applicable) agree that the interface/interconnection point and the metering point have been completed in accordance with the instructions of the Second Party.
- vi. All Parties agree to comply with all Government-issued provisions concerning plant operation and maintenance, drawings and diagrams, on-site duty schedule, harmonics, synchronization, voltage, frequency, flicker, etc.
- vii. In order to enable the Second Party to fulfill its obligation to maintain a safe and reliable distribution system, the First Party and, the Third Party (where applicable) agree that if the Second Party determines that the consumer's renewable energy system is causing damage to the Utility's assets or adversely impacting other consumers, the First/Third Party shall, upon instruction from the Second Party, disconnect the renewable energy system from the distribution grid and shall, at its own cost, undertake corrective measures to the satisfaction of the Second Party.
- viii. If, during a grid outage, back-feeding into the Utility's grid from electricity generated by the First/Third Party causes any accident resulting in injury or death (human or animal), whether fatal/non-fatal, departmental/non-departmental, the entire liability shall rest with the First/Third Party. In such exigencies, the Second Party reserves the right to disconnect the First/Third Party's system at any time.

3. Permits and Approvals

Where approvals and clearances from authorities other than the concerned Utility are required, the First/Third Party has obtained such approvals prior to connecting the photovoltaic system to the distribution system.

4. Access and Disconnection

- i. The Second Party shall have access at all times to metering equipment and to both automatic and manual means of disconnecting the renewable system.
- ii. In emergencies, where neither the automatic nor any manual disconnection device (e.g., switch or breaker) can be actuated, the Utility may disconnect the consumer's service connection at the premises.
- iii. In installations under the OPEX model, if the First Party fails to pay the Third Party under the OPEX agreement for two (2) consecutive months, and the Second Party verifies such default, the Second Party may disconnect the First Party's electricity connection.

5. Responsibility

- i. For any damage or adverse impact arising from negligence or willful misconduct of the First/Third Party in providing or operating the renewable energy connection, the First/Third Party shall compensate the Second Party, and the First Party shall ensure that.

- ii. The Second Party shall not be obligated to provide any financial or other incentives to the First Party beyond those stipulated in the Net Metering Guideline–2025, or any Government incentives expressly provided thereunder”.

6. Metering

Metering arrangements shall be as determined under Clause 3.6 and Annex 2 of the Net Metering Guideline–2025, including any amendments made from time to time.

7. Commercial Settlement/Arrangements

Commercial arrangements shall be executed in accordance with Clauses 3.4 and 3.5 of the Net Metering Guideline–2025, and any subsequent amendments.

Settlement Unit Price shall be paid by the Distribution Utility to the electricity consumer through the method selected below and, where applicable, to the account specified:

☐ Method-1: Bank Account (BEFTN/NPSB)	☐ Method-2: Mobile Banking Account (Nagad/bKash/Rocket)	☐ Method-3: Cheque
1. Account Name:	1. ☐ Nagad / ☐ bKash / ☐ Rocket	1. Account Name:
2. Account Number (13 digits):	2. Account Name:	
3. Bank Name:	3. Account Number:	
4. Branch Name:		
5. Routing Number:		

8. Cost of Providing Interconnection

All costs for the installation of the rooftop solar system, including metering and interconnection, have been borne by the First/Third Party. Where applicable, the First/Third Party has borne, and shall continue to bear, the capital cost of system modifications and the costs of system upgrades.

9. Post-Installation Inspection, Testing, Calibration, and Maintenance upon First Party’s Invitation

After installation, the Utility has carried out, upon the First Party’s invitation and through mutual scheduling with the consumer, all required inspections and tests. The First and Third Parties have provided the Second Party with all drawings, specifications, and test records of the project or generating station (as applicable), and shall provide such documents in the future as required.

10. Records

All Parties shall maintain complete and accurate records and any other data necessary for proper operation of the rooftop solar system under this Agreement. For OPEX model installations, a copy of the previously executed OPEX Agreement (as per Annex-6) between the First and Third Parties shall be attached hereto and shall form an integral part of this Agreement.

11. Dispute Resolution

- i. Any dispute between any two Parties arising out of or in connection with this Agreement shall first be attempted to be resolved amicably, promptly, fairly, and in good faith through mutual discussions.
- ii. If such dispute is not resolved through discussions within sixty (60) days from the date of its occurrence, or within an extended period mutually agreed by the Parties, then:
 - a. Disputes between the First and Third Parties regarding billing for energy export or the amount of billing shall be resolved by the Second Party (Utility).
 - b. Disputes between the First and Second Parties regarding billing for energy export or the amount of billing shall be resolved by the Power Division/SREDA.
 - c. Disputes relating to regulations or their interpretation shall be resolved in accordance with the decision of the Power Division; Ministry of Power, Energy and Mineral Resources, or by any agency authorized by the Power Division through due process. Such decision shall be considered final.

12. Contract Termination

- i. The First Party may, at any time, terminate this Agreement by giving the Second Party thirty (30) days' prior written notice.
- ii. The First/Third Party agrees that, upon any notice of breach issued by the Second Party, the First Party shall take necessary remedial action within thirty (30) days from receipt of such notice; failing which, upon expiry of thirty (30) days, the Second Party may terminate this Agreement by written notice.
- iii. Within one (1) week after termination of this Agreement, the First Party shall disconnect its renewable energy system from the Utility's distribution grid to the satisfaction of the Second Party.
- iv. The First/Third Party agrees that, if no electricity is generated from the renewable energy system installed under this Guideline for a period of one (1) year, or within any extended period granted by the Second Party, other than due to the Second Party's failure, the Second Party may terminate this Agreement by written notice.
- v. The First/Third Party agrees that, if any significant component of the system is replaced or newly added with any non-approved component in violation of the Guideline, the Second Party may terminate this Agreement by written notice.

Notification

For all communications under this Agreement, the following officers shall be responsible:

On behalf of the Utility	On behalf of the Electricity Consumer	On behalf of the OPEX Investor
---------------------------------	--	---

All Parties, acting through their duly authorized representatives, execute this Agreement for implementation purposes, and copies hereof existing on the above-stated date have been delivered to each Party.

FOR AND ON BEHALF OF The Utility	FOR AND ON BEHALF OF The Project Owner	FOR AND ON BEHALF OF The OPEX Investor
Authorized Signatory Witness 1.	Authorized Signatory Witness 1.	Authorized Signatory Witness 1.
2.	2.	2.

10 Annex 5: Examples of Energy Accounting and Billing Process in Net Metering

a. When, considering import, export and credit units during a monthly billing period, the Billing Unit becomes zero

Assume Mr. Abdul Karim is a Bangladesh Power Development Board (BPDB) consumer under MT-3: Industrial. His Sanctioned Load is 500 kW. He installed a 500 kW AC rooftop solar system under net metering.

In November 2024, he imported (I) 50,000 units from the grid and, after self-consumption, exported (E) 48,000 units to the grid. Since export is less than import, his net export is $E_{net} = 0$, hence no maintenance charge applies. His previous month (October 2024) credit was 2,000 units ($C = 2,000$).

Therefore, his Billing Unit (BU) is: $BU = (I - E - C) = (50,000 - 48,000 - 2,000) = 0$ units.

As import exceeds export, no credit remains at month-end. Mr. Karim's bill for November 2024 is as follows:

Description of electricity export/ import	Amount	Rate (in BDT)	Total bill (in BDT)
Demand Charge	500 kW	BDT 90/kW/month	45,000
Units imported from grid (I)	50,000 units	—	—
Units exported to grid (E)	48,000 units	—	—
Net export (E_{net})	0 units	—	—
Adjustable export (E_{adj})	0 units	—	—
Previous month's credit (C)	2,000 units	—	—
Net Billing Units (BU)	0 units	—	—
Credit carried forward (C)	0 units	—	—
Subtotal	—	—	45,000
VAT on subtotal	—	5%	2,250
Total bill (excl. late fee & meter charge)	—	—	47,250

Because Abdul karim's monthly import is exactly offset by current-month export plus prior credit, so he need not pay the energy charge in this month—only other fixed charges need to be paid.

b. When, considering import, export and credit units during a monthly billing period, the consumer is a net exporter):

Assume Mr. Abdus Salam is a DESCO consumer under LT-E: Commercial & Office with Sanctioned Load = 20 kW and a 20 kW AC rooftop solar system. In November 2024, he imported 500 units off-peak ($I_{\text{off-peak}}$) and 1,000 units peak (I_{peak}), so $I = 1,500$ units. He exported $E = 2,000$ units. His previous month's credit was $C = 200$ units. Thus, $E_{\text{net}} = (E - I_{\text{off-peak}} - I_{\text{peak}}) = (2,000 - 1,000 - 500) = 500$ units. After deducting 10% system maintenance from export energy (E_{net}), Mr. Abdul Salam's the amount of adjustable export energy: $E_{\text{adj}} = 0.9 \times 500 = 450$ units. Credit to carry forward next month (December 2024) is $E_{\text{adj}} + C = 450 + 200 = 650$ units.

Mr. Abdul Karim's bill for the month of October, 2018 is shown below:

Import/Export Description	Off-peak ($I_{\text{off-peak}}$)	Peak (I_{peak})	Rate	Total (BDT)
Demand Charge	20 kW	—	BDT 90/kW/month	1,800
Imported units (I)	500	1,000	—	—
Exported units (E)	2,000	—	—	—
Net export (E_{net})	500	—	—	—
Adjustable export (E_{adj})	450	—	—	—
Previous credit (C)	200	—	—	—
Net Billing Units (BU)	0	0	$0 \times \text{tariff}$	0
Credit carried forward (C)	650	—	—	—
Subtotal	—	—	—	1,800
VAT	—	—	5%	90
Total (excl. late fee & meter charge)	—	—	—	1,890

After all the electricity imported by Mr. Abdus Salam this month has been offset against his exported electricity, the remaining 650 units are retained as an adjustable credit, which will be applied in the following month's adjustment. He is not required to pay any energy charge for this month and will only need to pay the applicable fixed charges.

- c. **When, considering import, export and credit units during a monthly billing period, the consumer is a net importer):**

Let us consider that Ms. Rafia Begum is a residential electricity consumer under a Rural Electrification Cooperative of the Bangladesh Rural Electrification Board (BREB). Her sanctioned load is 10 kW. To avail of the net metering facility, she has installed a rooftop solar system with a capacity of 10 kW AC. In April 2024, Ms. Rafia Begum imported (consumed) 500 units (I) of electricity from the grid. During the same month, after meeting her own consumption, she exported 350 units (E) of electricity to the grid from her solar system. Since her imported electricity exceeds her exported electricity, there is **no net export**, i.e., $E_{net} = 0$. Therefore, she is not required to pay any maintenance charge. As there was no credit carried forward from the previous month (March 2024), $C = 0$. Hence, her **billing units (BU)** for April 2024 are calculated as: $BU = (I - E - C) = (500 - 350 - 0) = 150$ **units**. Since her exported electricity is less than the imported amount, she will have **no credit units carried forward** to the following month.

LT-A Residential slab tariffs:

Lifeline	0–50 units:	BDT 4.63
Step 1	0–75 units	BDT 5.26
Step 2	76–200 units	BDT 7.20
Step 3	201–300 units	BDT 7.59
Step 4	301–400 units	BDT 8.02
Step 5	401–600 units	BDT 12.67
Step 6	>600 units	BDT 14.61

The electricity bill for Ms. Rafia Begum for the month of April 2024 is shown below:

	Bill after installation of the solar system			Bill for the same consumption before installation of the solar system		
Description of Electricity Import/Export	Qty	Rate (BDT)	Total (BDT)	Qty	Rate (BDT)	Total (BDT)
Demand Charge ()	10 kW	BDT 42/kW/mo	420	10 kW	BDT 42/kW/mo	420
Imported units (I)	500	–	–	500	–	–

Exported units (E)	350	–	–	–	–	–
Net export (E _{net})	0	–	–	–	–	–
Adjustable export (E _{adj})	0	–	–	–	–	–
Previous credit (C)	0	–	–	–	–	–
Net Billing Units (BU)	150	–	–	500 unit	–	–
Credit carried forward	0	–	–	–	–	–
Step 1 (0–75)	75 unit	5.26	395	75 unit	5.26	395
Step 2 (76–200)	75 unit	7.20	540	125 unit	7.20	900
Step 3 (201–300)	–	–	–	100 unit	7.59	759
Step 4 (301–400)	–	–	–	100 unit	8.02	802
Step 5 (401–600)	–	–	–	100 unit	12.67	1,267
Step 6 (>600)	–	–	–	–	–	–
Subtotal	150 unit	–	1,355	500 unit	–	4,543
VAT (5%)	–	–	67.75	–	–	227
Total (excl. late fee)	–	–	1,422.75	–	–	4,770

Since the total of Ms. Rafia Begum's exported electricity and the previous month's credit units is less than her imported electricity, no energy credit will be carried forward. However, whereas she previously paid electricity charges up to the fifth slab, she now pays only up to the second slab due to the installation of the solar system. As a result, her current electricity bill amounts to Tk. 1,422.75, which is Tk. 3,347.25 less than her previous bill (Tk. 4,770 – Tk. 1,422.75 = Tk. 3,347.25). In other words, by installing the rooftop solar system, she has saved Tk. 3,347.25, representing approximately 70% reduction in her electricity expenses.

- d. When a prosumer is a net exporter of electricity, the sample procedure for preparing the electricity bill at the end of the settlement period (every three months) is shown below:

Ms. Aditi Chowdhury is an electricity consumer under the Dhaka Power Distribution Company (DPDC) in the LT-C1: Small Industry category with a sanctioned load of 50 kW. To avail of the net-metering facility, she has installed a 50 kW (AC) rooftop solar system. In June 2024, she imported 4,500 units (I) of electricity from the grid and exported 4,000 units (E) of electricity from her solar system after self-consumption. Since her imported electricity exceeds her exported amount, there is no net export ($E_{\text{net}} = 0$), and thus no maintenance charge applies. From the previous month (May 2024), she had a credit of 2,000 units ($C = 2,000$). Therefore, her billing units for June 2024 are calculated as $BU = (I - E - C) = (4,500 - 4,000 - 2,000) = -1,500$ units. The positive value of this difference, 1,500 units, will be treated as the settled or exported electricity units.

Description of electricity export/ import	Amount	Rate (in BDT)	Total bill (in BDT)
Demand Charge	50 kW	BDT 48/kW/month	2,400
Imported units in June (I)	4,500	—	—
Exported units in June (E)	4,000	—	—
Net export in June (E_{net})	0	—	—
Adjustable export (E_{adj})	0	—	—
Previous month's credit (May 2024) (C)	2,000	—	—
Settled units	1,500	BDT 8.56/unit*	-12,840
Credit for next month (C)	0	—	—
Subtotal	—	—	-10,440
VAT (5%)	—	—	522
Net payable/receivable	—	—	-9,918

* Bulk rate of DPDC: 8.56 BDT/unit

At the end of the settlement period, the distribution utility pays the prosumer the value of settled units (BDT 12,840) minus other fixed charges and VAT; the net receivable here is BDT 9,918, paid via the preset method (bank/MFS/cheque).

Note: At the close of each settlement period (March, June, September, December), if a contracted prosumer is a net exporter, the distribution utility shall settle payment at the BERC-determined bulk rate in accordance with the Guideline.

11 Annex 6: Important Issues about the Agreement between Consumer and OPEX Investor Responsible for System Installation in the OPEX Model

1. The consumer shall agree—per Net Metering Guideline-2025—to purchase all electricity generated by the solar system from the RE Accounting Meter/Solar Accounting Meter(s) at the price and terms stated in the OPEX agreement (generally lower than the distribution utility’s retail tariff). The consumer shall determine the tariff and select the OPEX investor through the appropriate process.
2. The tariff structure (flat or year-wise slabs) must be clearly stated in the agreement, including clear change conditions to determine future tariffs unambiguously during the contract term.
3. Payments shall be made via bank or comparable reliable channels with proper evidence. The agreement must state account details (account name, number, bank, branch, routing number) or equivalent.
4. Monthly meter readings of all meters shall be taken on time in the presence of representatives from both Parties and preserved in records by both.
5. By mutual consent, the consumer shall provide a standby LC guarantee (or similar security) covering at least 12 months; by mutual consent and with explicit contractual provision, this may be waived—but the Third Party may face future risk.
6. If the consumer’s annual settled units are not zero, the adjustment terms must be specified clearly in the OPEX agreement.
7. The quantity and conditions for drawing reactive power from the solar system shall be included in the agreement.
8. Responsibilities assigned jointly to the consumer and the OPEX investor under Annex 4 and the Guideline must be apportioned clearly in the OPEX agreement.
9. The OPEX agreement shall have a fixed term, and a regular exit plan for the solar system at end-of-term shall be specified.
10. For forced early termination at the consumer’s proposal, the Parties’ actions shall be specified; similarly, for forced early termination at the OPEX investor’s proposal.
11. Subject to mutual consent and a specified fee, the distribution utility may collect OPEX bills from the consumer on behalf of the OPEX investor. This provision must be included, and a copy of the utility-OPEX collection agreement or the utility’s consent letter attached. This fee may include the national grid operator’s grid system maintenance charge.

12. A copy of this bilateral OPEX agreement shall, after net-metering system installation, be attached to the tripartite agreement under Annex 4 and deemed an integral part thereof.

12 Annex 7: Billing Format for Net-Metered Consumers as Published by the Commission

The current billing format as published by the Commission is provided hereunder and shall be subject to revision by the Commission from time to time.

		ORGANIZATION NAME Office of the..... NET METERING BILL							
Consumer Name:		MONTH		BILL NO		CD		ISSUE DATE	
Address:		LOCATION		BILL GROUP		BOOK NO		WALK ORD	
RID:		PRV A/C NO		CONSUMER NO		DUE DATE			
E-mail:		TARIFF		Status		CT		PT	
Mobile:		S.LOAD (kW)		Max.Demand (kW)		Solar Capacity(kW)		Mode	
MTR TYPE		METER NO		COND		OMF		Meter Seal No.	
IMPORT-EXPORT									
CARRY OVER UNIT		Import Meter		Export Meter					
READING		kWh		kVarh		kWh			
DETAIL		DATE		SIR.BRS./OFF PEAK		PEAK		SIR.BRS./OFF PEAK	
PRESENT									
PREVIOUS									
DIFFERENCE									
CONSUMED (001.9 H 000)									
ADJUSTED UNIT									
x-FORMER LOSS UNIT									
OLD METER UNIT									
SUB M.CONSUM.									
NET BILLED UNIT				PF		CARRY FORWARD UNIT			
DEMAND METER READING DETAIL (kW)		PRESENT		PREVIOUS		CONSUMED		SETTLEMENT UNIT	
								SETTLEMENT RATE	
CURRENT CHARGES		TAKA		OTHER CHARGES		TAKA			
S.REG.ELEC.CHARGE				ADJ.PRINCIPAL					
OFF PEAK ELEC.CHARGE				ADJ.L.P.S					
PEAK ELEC.CHARGE				ADJ.VAT					
DEMAND CHARGE				ADV/ARP.PRINCIPAL					
METER RENT				CURR.& ARR.L.P.S					
PF SURCHARGE				ADD./ARR.VAT					
MISC BILL/FEE:				NEM SETTLEMENT AMT.					
				PRINCIPAL SETTLEMENT					
				VAT SETTLEMENT					
PRINCIPAL AMOUNT				VAT TOTAL					
VAT				AMOUNT TO BE PAID					
Bill Month total				BY CONSUMER					
LAST PAYMENT		[DATE]		[AMOUNT]		AMOUNT TO BE PAID			
						TO CONSUMER			
MONTH		PRINCIPAL		MISC BILL/FEE:		LPS		VAT	
								TOTAL (TK)	
PAY AT									
Office of the.....								BANK & OFFICE COPY	
SERIAL NO		LOCATION		BILLGR		BOOK NO		WALK ORD	
RCVD TK.									
DATE		CONSUMER NO		ISSUE DATE		DUE DATE			
TARIFF									
		BILL NO		CD		VAT (TK)		TOTAL AMOUNT (TK)	
		BILL MONTH		LPS		AMOUNT TO BE PAID TO CONSUMER			
CONS NO									

13 Annex 8: Implementation of Net Metering on Pre-payment/Smart Meters

1. The pre-payment/smart meter and/or its server system must be made bidirectional, incorporating, in addition to existing parameters, the capability to record and store data required by net metering, including Export, Import, Credit, Settlement Unit, Settlement BDT Amount, etc. Historical data for Export, Import, Credit, Settlement Unit must be stored by settlement period.
2. At the end of each settlement month (end of March, June, September, December), the accumulated credit units shall be treated as settlement units, and credit shall be reset to zero going forward. If settlement units are non-zero, the distribution utility shall provide the consumer a detailed billing document per Annex 7 and pay the value of settlement units via the specified method (bank/MFS/cheque).
3. Each month, the distribution utility shall notify the consumer via SMS/email/mobile app/web portal and display on the meter: import, export, net export, previous credit, current credit, settlement units and their value. Total import and export readings must be recorded and displayed on the meter.
4. At the end of each billing period, all relevant registers shall be updated per the Guideline after computing import units, export units, net export units, distribution system maintenance charge, prior credit units, etc. Any consumer credit units must be stored in the credit register.
5. Against consumer usage, adjustment shall first be made from credit units; once credit reaches zero, deductions shall then occur from the recharged balance as per rules. No deduction may be made from the recharged balance while credit units remain.
6. For slabbed consumers, slab calculation shall be based only on the units for which the recharged balance is deducted.
7. Smart pre-paid meters used for net metering shall be re-programmable. If in future BERC applies a separate export tariff, the same meter must support separate tariffs for import and export.
8. In all cases (pre-payment/post-paid/smart), the distribution utility shall share import and export data of each meter from the meter server with SREDA through an automated online system. SREDA may verify such data as required.
9. New smart meters must incorporate the above provisions to enable activation by the utility as needed. For existing meters, the utility shall enable these features through re-programming as required.

14 Annex 9: Verification Committee for Installing a Renewable System with Capacity Exceeding the Contracted Load

If a consumer wishes to generate from renewable sources more than the contracted load, they must submit a separate application to the relevant distribution entity. If deemed appropriate, the distribution entity shall forward the case to the Grid Stability Committee for opinion. The Power Division will decide based on the Committee's recommendation. The outline of the Committee is as follows:

- | | |
|--|------------------|
| 1. Executive Engineer, NOD, LDC, PGCB PLC - | Convener |
| 2. Representative, Bangladesh Power Development Board - | Member |
| 3. Representative, Bangladesh Rural Electrification Board - | Member |
| 4. Representative, Dhaka Power Distribution Company Ltd. - | Member |
| 5. Representative, Dhaka Electric Supply Company Ltd. - | Member |
| 6. Representative, West Zone Power Distribution Company Ltd. - | Member |
| 7. Representative, Northern Electricity Supply PLC - | Member |
| 8. Executive Engineer, IMD, LDC, PGCB PLC - | Member-Secretary |